Challenges Faced

- Setting up and configuring Apache Spark, PySpark, and HDFS correctly was challenging.
- Handling the large 10-million-record dataset required careful resource and execution management.
- We faced dependency issues such as PySpark and Python package configuration inside the virtual environment.
- Selecting and preprocessing the correct customer features for clustering required attention.
- Running multiple K-Means models for different k values increased execution time.
- Interpreting WSSSE and the Elbow Curve correctly was another challenge.
- Comparing single-node K-Means with Spark K-Means showed the impact of distributed processing overhead.